INCH-POUND

MIL-DTL-83503/25C
21 July 2004
SUPERSEDING
MIL-DTL-83503/25B
17 June 1988

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL FOUR WALL HEADER, STRAIGHT THRU (.100 SPACING) WITH POLARIZING FEATURE, SOLDERLESS WRAPPOST OR PWB TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83503.

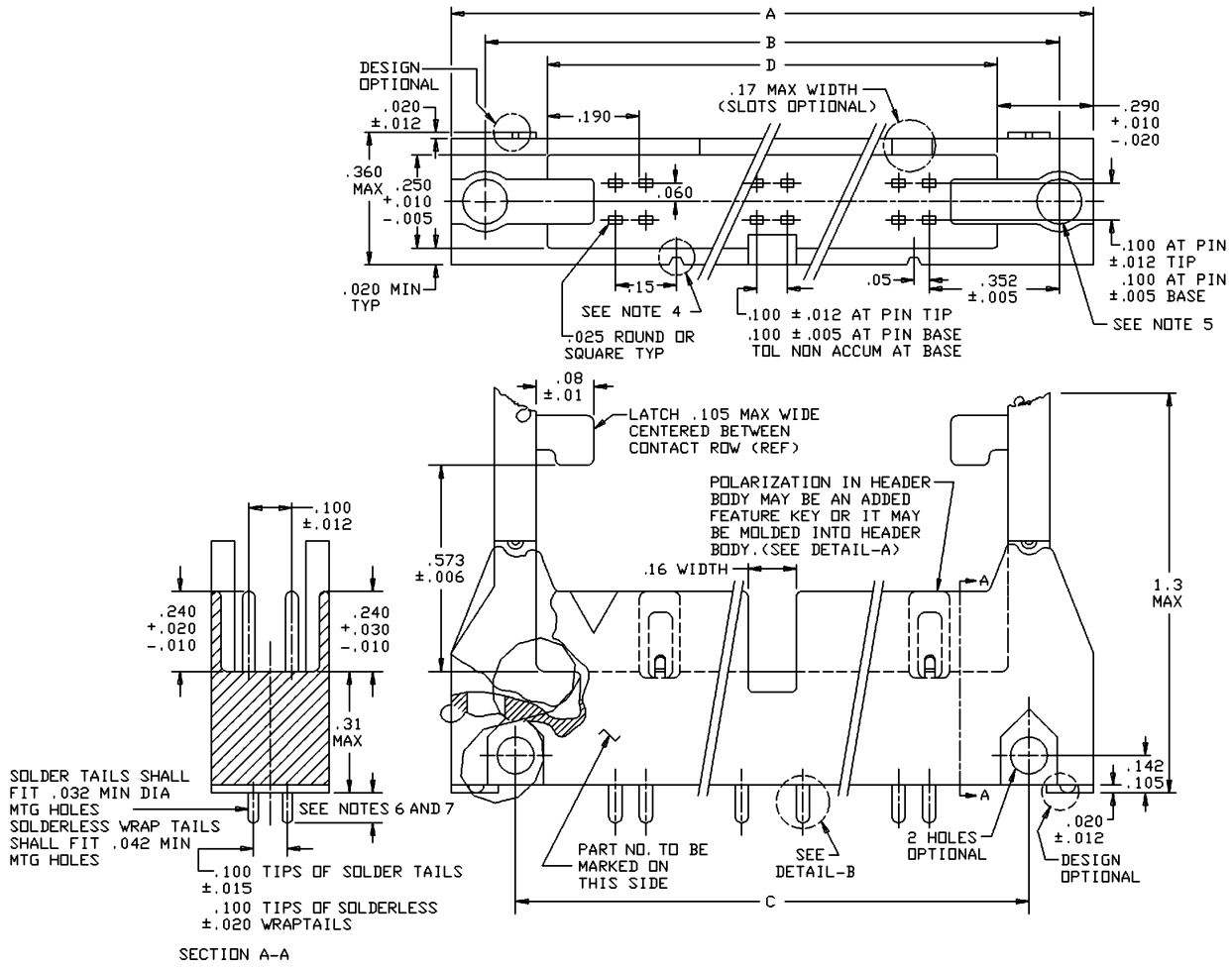


FIGURE 1. Dimensions and configuration.

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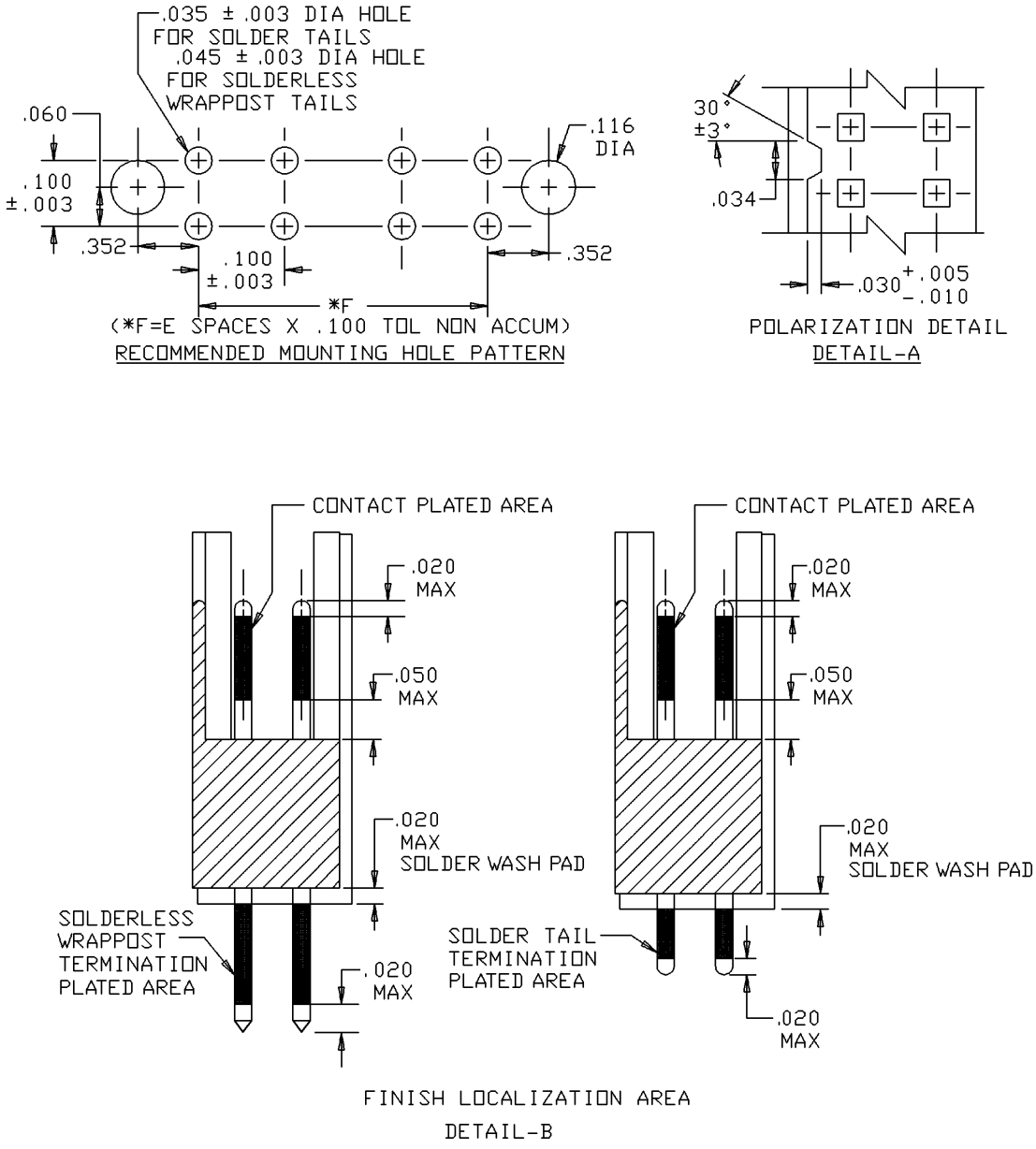


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.030	0.76	.08	2.0	.17	4.3
.005	0.13	.032	0.81	.089	2.26	.178	4.52
.006	0.15	.034	0.86	.100	2.54	.190	4.83
.010	0.25	.035	0.89	.105	2.67	.240	6.10
.01	0.3	.042	1.07	.116	2.95	.250	6.35
.012	0.30	.045	1.14	.125	3.18	.290	7.37
.015	0.38	.05	1.3	.142	3.61	.31	7.9
.020	0.51	.050	1.27	.15	3.8	.352	8.94
.025	0.64	.060	1.52	.152	3.86	.360	9.14
		.062	1.57	.16	4.06	1.3	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ± .005 inch (0.13 mm) for three place decimals and ± .01 inch (0.3 mm) for two place decimals.
4. For 10 and 14 position connectors the polarizing feature on this end is non-existent.
5. Two thread cutting screws, No. 4-24 (1/4 inch (6.35 mm) length) type BG are provided.
6. Pin length for printed wiring board (PWB) shall be as follows:

PWB thickness inch (mm)	Pin length inch (mm) min	Pin length inch (mm) max
.062 (1.57)	.089 (2.26)	.116 (2.95)
.125 (3.18)	.152 (3.86)	.178 (4.52)

7. Solderless wrappost, pin length shall be .610 ± .015 inch (15.49 ±0.38 mm).

FIGURE 1. Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	Number of pins	Dimensions						Termination, right angle contact type
		A inch (mm)	B inch (mm)	C inch (mm)	D 1/ (mm) +.030 (0.76) -.000	E spaces	F inch (mm)	
-01 -02 -03	10 10 10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	.062 PWB .125 PWB Wrappost
-04 -05 -06	14 14 14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.062 PWB .125 PWB Wrappost
-07 -08 -09	16 16 16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.062 PWB .125 PWB Wrappost
-10 -11 -12	20 20 20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.062 PWB .125 PWB Wrappost
-13 -14 -15	26 26 26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48	.062 PWB .125 PWB Wrappost
-16 -17 -18	34 34 34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.062 PWB .125 PWB Wrappost
-19 -20 -21	40 40 40	2.76 (70.1	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.062 PWB .125 PWB Wrappost
-22 -23 -24	50 50 50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	.062 PWB .125 PWB Wrappost
-25 -26 -27	60 60 60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	.062 PWB .125 PWB Wrappost
-28 -29 -30	64 64 64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	.062 PWB .125 PWB Wrappost

1/ Tolerance for “D” dimension is +.030 -.000 inch (+0.76 -0.00 mm).

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° to +120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Contacts shall be phosphor bronze, copper alloy, or brass. Phosphor bronze in accordance with ASTM B139, ASTM B159 or ASTM B103. Copper alloy UNS No. C51000 or brass in accordance with copper alloy CDA 260 spring temper; or copper alloy UNS No. C72500 spring temper in accordance with ASTM B122.

Contact plating:

Contact area: Gold in accordance with MIL-DTL-83503.

Solder tail termination area: Gold, or tin-lead, in accordance with MIL-DTL-83503.

Housing: Glass field polyester in accordance with MIL-M-24519, type GPT-15F, GPT-20F, GPT-30F.

Accessories: Parts may be constructed of polyester in accordance with MIL-M-24519 or other materials specified in MIL-DTL-83503, provided these parts do not come in contact with current carrying components.

Polarization key: Shall be permanently attached to header (added feature key shall be bonded to header).

Mating connector: See MIL-DTL-83503/7.

Solderless wrappost contacts:

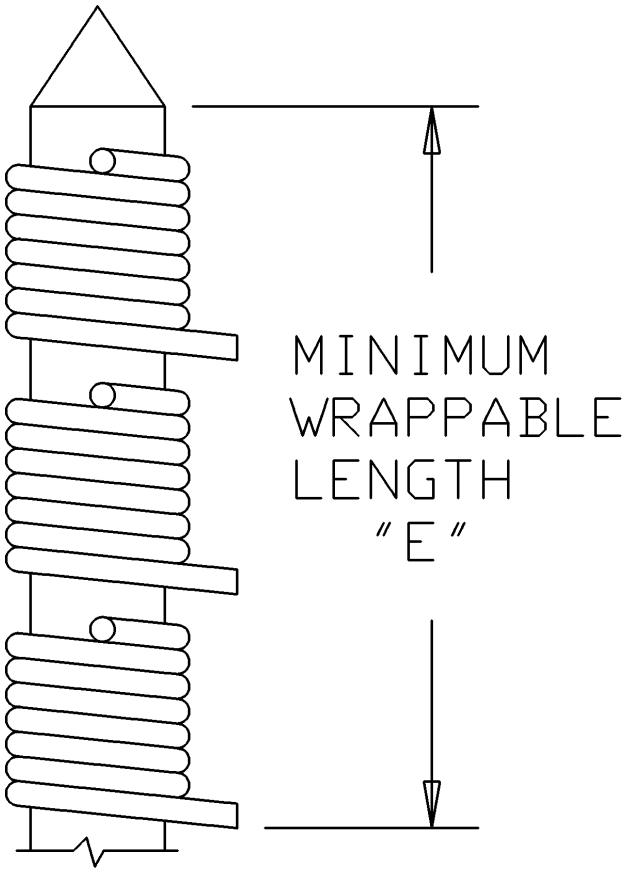
Wrappost geometry: .045 inch (1.14 mm) square wrappost in accordance with figures 2 and 3.

Parallelism: See figure 3 and table.

Tip configuration: The tip configuration of the wrappost shall terminate in a radius or bevel to facilitate insertion of the wrapping tool. See figure 3.

Solderless wrappost contact wire accommodation: 26 or 28 AWG solid.

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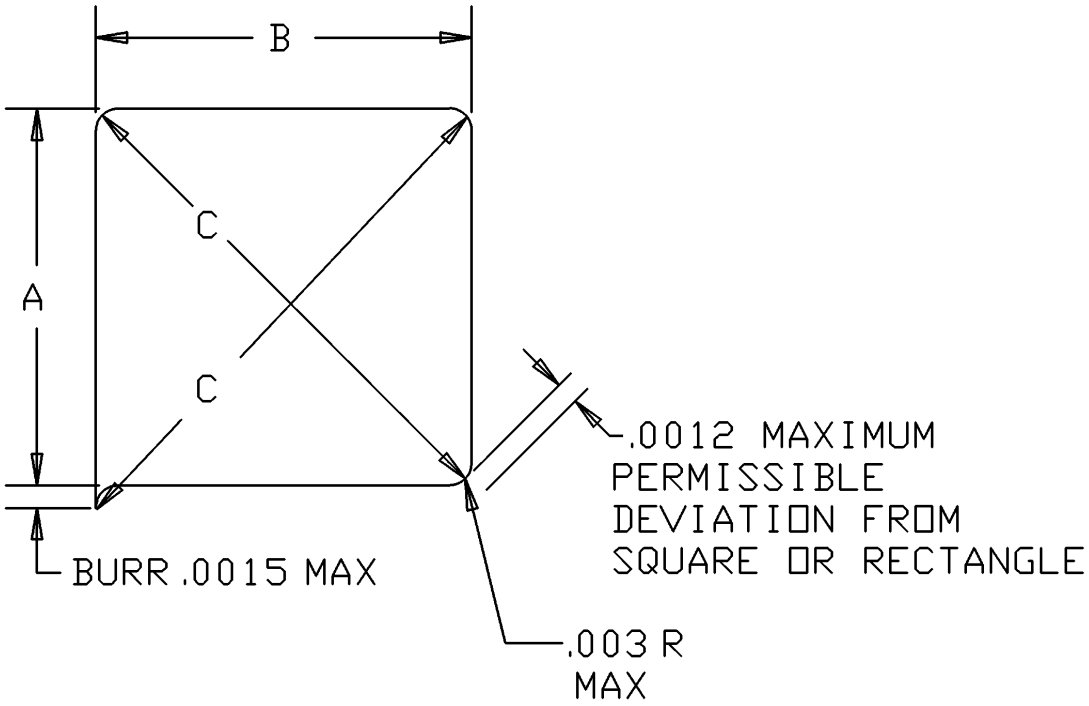


Number of wrapped corrections	"E" Length	
	Wire gage 28 inch (mm)	Wire gage 26 inch (mm)
1	.219 (5.56)	.226 (5.74)
2	.388 (9.85)	.402 (10.21)
3	.557 (14.15)	.578 (14.68)

NOTES:
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2. Metric equivalents are given for information only.

FIGURE 2. Wrappable length (typical connection).

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Inches	mm
.0012	0.030
.0015	0.038
.003	0.076

A inch (mm)	B inch (mm)	C Inch (mm)	Parallelism inch (mm)	Straightness Inch/inch
.025 (0.64) nominal .022 (0.56) min	.025 (0.64) nominal .022 (0.56) min	.0355 (0.902) max .0325 (0.823) min	.002 (0.05)	.005

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. If the tip of the wrappost terminates in a bevel, the apex of the bevel shall be flat, with no side of the flat exceeding .015 inch (0.38 mm).

FIGURE 3. Wrappost geometry.

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Withstanding voltage

Sea level: 500 V rms, minimum.

Altitude: 200 V rms, minimum.

Cable retention: Not applicable.

Contact retention. Not applicable

Connector-cable flexing: Not applicable.

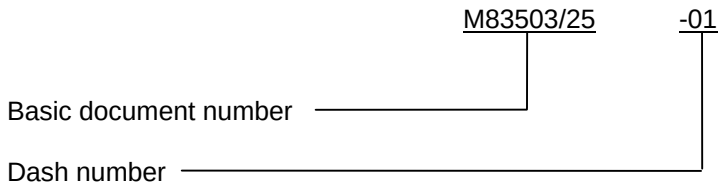
Contact resistance: 50 milliohms maximum.

Latch retention: The latch shall retain the mating connector when the cable retention force is applied to the mating connector cable. The latch shall also function as an ejector to eject mating connector.

Marking:

Part or Identifying Number (PIN): The PIN shall consist of the letter “M”, followed by the basic specification number, “/”, the specification sheet number, “-” and dash number from table I.

Example:



Contact identification: Molded triangle “ Δ ” or notch to show number 1 contact, or raised numbers to show position number “1”.

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Referenced documents: In addition to MIL-DTL-83503, this document references the following:

MIL-DTL-83503/7
MIL-M-24519
ASTM B103
ASTM B122
ASTM B139
ASTM B159
CDA 260

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4601-004)

Review activities:
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

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